CHAPTER 3

Innocent (Commutative) Algebra

"The introduction of the cipher 0 or the group concept was general nonsense too, and mathematics was more or less stagnating for thousands of years because nobody was around to take such childish steps..."
— Alexander Grothendieck (in his 1982 letter to Ronald Brown)

"Taking a new step, uttering a new word, is what people fear most."
— Fyodor Dostoevsky, Crime and Punishment

3.1. It will not be possible for us to discuss the complete (basic) algebra. We will assume some knowledge and this chapter should be treated to recall some of the basic facts. If one is not comfortable with algebra of rings, fields, vector spaces (and modules), then one may refer to [**1**, **2**]. Even though the chapter is set for algebra, we take some irregular detours to other parts of mathematics.

1. Some Preliminaries

Definition 3.2 (Subset). For two sets S and T we say that S is a *subset* of T if each element of S is also an element of T, i.e. if S $\subseteq$ T $\forall\ x \in S$ we have $x \in T$.
Simultaneously, we can also say that T contains S which can be written as $T \supseteq S$. If $S \subseteq T$ and $S \neq T$ hold, then we write $S \subset T$ and we say that S is a *proper subset* of T.

Remark 3.3. Two sets are equal iff each is a subset of the other. In symbolic notation: $(A = B) \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

Definition 3.4 (Cartesian Product). Consider a given family of sets $(S_i)_{i \in I)}$, where I denotes the index set, then the cartesian product denoted as $\prod_i S_i$, of the family of given sets is the set of all functions f from the index set I with f_j in each S_j, for each $j \in I$.

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For a finite family of sets, we can define the Cartesian Product as:

$$\prod_{i=1}^{i=n} S_i = \{(s_1, s_2, s_3, \cdots, s_n) \ \mid \ s_i \in S_i, \ \forall i \in \{1, 2, 3, \cdots, n\}\}$$

Definition 3.5 (Relation). A relation is a subset of the cartesian product of two sets that helps to establish a connection between the elements of the sets. Given a set X, a relation R over X is a set of ordered pairs of elements from X:

$$\mathbb{R} \subseteq \{(x, y) \mid x, y \in X\} \tag{1}$$

Definition 3.6 (Injection, Surjection and Bijection). A function $f : S \hookrightarrow T$ is *injective* or one-to-one if no element of T (no second coordinate) appears in more than one ordered pair. Such a function is called an **injection**. A function $f : S \twoheadrightarrow T$ is *surjective* or onto if every element of T appears in an ordered pair. Such a function is called a **surjection**.

A function that is both surjective and injective is said to be *bijective*. Bijective functions are called **bijections**.

3.7. One should be careful that all of these definitions are apt for set theoretic foundations only. We would revise most of these definitions in future sections, especially in category theory.

Definition 3.8 (Equivalence Relation). A binary relation[1] $\sim$ on a set is said to be an equivalence relation, iff it is *reflexive, symmetric* and *transitive*. An equivalence class of an element a is defined as:

$$[a] = \{x \in X : \ a \sim x\}. \tag{2}$$

The set of all *equivalence classes* in X with respect to an equivalence relation R is denoted as X/R and is called X modulo R.

3.9. Let us write the axioms[2] for reflexive, irreflexive, symmetric, anti-symmetric, asymmetry and transitive relations.

Reflexive: $\forall a, \ a \sim a$
Symmetric: $\forall a, b, \ a \sim b \implies b \sim a$
Transitive: $\forall a, b, c, \ a \sim b$ and $b \sim c \implies a \sim c$
Asymmetry: $\forall a, b, \ a \sim b \implies \neg\, b \sim a$
Anti-symmetry: $\forall a, b, \ a \sim b, \ b \sim a \implies a = b$
Irreflexive: $\forall a, \ \neg a \sim a$ (for example $(\mathbb{R}, <)$)

[1]In formal logic, a *predicate* or relation is an n-ary symbol and so is an operation. Binary operation is just an example of n-ary operations.

[2]Given any relation, there are some properties that we list down which are called axioms. For instance, axioms of $(\mathbb{R}, \leq)$ consists of reflexivity, anti-symmetry, and transitivity.

REMARK 3.10. The symbol '$\neg$' denotes the negation statement.

DEFINITION 3.11 (Congruence Relation). If a and b have the same remainder when they are divided by a non-zero integer m, then we say that *a is congruent to b mod m*, represented as:

$$a \equiv b(mod\ m) \tag{3}$$

Congruence modulo m is an equivalence relation which is compatible with the operations of addition, subtraction, and multiplication.

DEFINITION 3.12 (Residue Class). The residue class of a mod m is denoted as $[a]$, where $[a]$ represents the set of all integers that are congruent to a modulo m.

DEFINITION 3.13 (Partial Order and Total Order). The reflexive, antisymmetric and transitive relation $\mathcal{R}$ is called a partial order relation and $\mathcal{R}$ is said to define a partial ordering of $\mathcal{S}$. The addition of the trichotomy law to the set of axioms of the partial order, defines the total order.

Hence, for a set $\mathcal{S}$ to be totally(/partially) ordered, it must have a relation $\mathcal{R}$ on it with the total(/partial) order.

EXAMPLE 3.14. $(\mathbb{R}, <)$ is an example of strict linear order which is totally ordered and $(\mathbb{R}, \leq)$ is an example of linear order which is partially ordered.

LEMMA 3.15 (Kuratowski-Zorn Lemma a.k.a. Zorn's Lemma). If every totally ordered subset of a poset $\mathcal{S}$ has an upper bound, then $\mathcal{S}$ contains a maximal element.

2. Algebraic Structures

DEFINITION 3.16 (Algebraic Structure). An algebraic structure is defined as a non-empty set that forms the closure with at least one operation, denoted as $(S, \star)$, where S denotes the non-empty set and $\star$ defines the operation.

In accordance with the above definition, any non-empty set with an operation will form an algebraic structure and with the duality principle, will also possess a geometrical interpretation.

DEFINITION 3.17 (Semigroup, Monoid, Groupoid). A *semigroup* is an algebraic structure equipped with a binary operation and satisfying associativity.

A *monoid* is a semigroup fulfilling the existence of a unique identity.

A *groupoid* is a monoid consisting of an inverse map with respect to every map that exists.

3.18. A group can be seen as a set of *symmetries* of any object. And symmetries, on the other hand, will be defined as mapping an entity to itself, while preserving the structure.

Definition 3.19 (Group). A group is a set G with a binary operation $G \times G \Rightarrow G$, usually written as

$$(a, b) \Rightarrow a + b, a \times b, ab, \ or \ a \cdot b, \tag{4}$$

and must follow the following *axioms*:

(1) $\exists$ a unique identity $(e, 1, \ or \ 0)$ such that $ea = a = ae$
(2) $\exists$ a 2-sided inverse, for every element a^{-1} such that $a^{-1}a = aa^{-1} = e$
(3) $\exists$ associativity amongst the elements such that $(ab)c = a(bc)$

3.20. Alternatively, we can also see a group as a monoid, consisting of a 2-sided inverse of all the elements in the underlying set.

Definition 3.21 (Commutative Group). Let $(G, \star)$ be a group, then G is said to be a commutative group if $a \star b = b \star a, \ \forall \ a, b \in G$.

3.22. We say that a group structure $(G, \star)$ is **Abelian** if it is commutative. That is, in an Abelian Group, the order of applying the group operation on the elements becomes irrelevant. Hence, the elements have become invariant under the order of application of the group operation.

Example 3.23 (Example of Finite Abelian Group). A group G defined as $(G = \{e, a, b, c\}, \star)$ [3] , where $\star : G \times G \longrightarrow G$, such that each element is self inverse and when two different elements other than identity are multiplied, then the third element, other than the identity is the result.

Definition 3.24 (Order of an Element). For a group G and $x \in G$, the order of x is defined as the smallest positive integer n such that $x^n = e$, where e is the identity of G. It is denoted as $o(x)$.

If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of infinite order.

3.25. An element of a group has order 1 iff it is the identity.

Definition 3.26 (Order of a Group). For a discrete group G, the **order** of the group is the cardinality of the underlying set. Hence, for a finite group G, its order $|G|$ is a natural number.

3.27. The *order of a subgroup* is similar to a group, i.e., it represents the number of the elements present in the subgroup.

Theorem 3.28 (Fundamental theorem of order). Let G be a group and $g \in G$ be an element then,

[3] Klein four-group

(1) If $|g| = \infty$, then $g^n = g^m$ iff $n = m$.
(2) If $|g| = k$ is finite, then $g^n = g^m$ iff $k|(n - m)$.

Definition 3.29 (Group Generator). A group generator is a subset of a group, such that every element of the group can be expressed as a finite combination of elements and their inverses. This can be expressed notationally as:

$$G = \langle S \rangle \tag{5}$$

and we say that G *is generated by* S or S *generates* G.

Definition 3.30 (Relations). Any equations in a general group G that the generators satisfy are called **relations** in G.

Example 3.31. For the group $\mathbb{Z}$, the set of all integers, both -1 and 1 are generators, as every integer number can be reached by repeatedly adding or subtracting these values.
Similarly, the cyclic group Z_n consists of integers modulo n, and a single generator can generate it, namely 1 or -1.

Example 3.32. In the complex numbers group $\mathcal{C}$, under multiplication, the unit complex numbers $e^{i\theta}$ where θ is a real number, can generate all the other elements.
Since, any complex number can be written in the polar form $re^{i\theta}$, where r is a real number representing the magnitude and θ is an argument.

Definition 3.33 (Subgroup). Consider a group G, and a subset H of G. H is a subgroup of g, iff H also froms a group under the same operations as in G.

3.34. S generates G iff the smallest subgroup of G containing S is G itself (the improper subgroup).

- A *proper subgroup* is the subgroup H which is the proper subset of G, i.e., $G \neq H$, represented as, $H < G$.
- Since $G \subseteq G \Rightarrow (H = G, \star)$ is also a subgroup of G, referred to as the *improper subgroup* of $(G, \star)$.
- A *trivial subgroup* is the subgroup $\{e\}$, consisting of just the identity element.
- If H is a subgroup, then G would be referred to as the *overgroup* of H.

Definition 3.35 (Relations). Any equations in a general group G that the generators satisfy are called relations in G.

3.36. To check if a set H equipped with a binary operation forms a subgroup of a group G, one needs to check for the presence of the identity elements, closure under the law of composition and for the existence of inverses.

Definition 3.37 (Normalizer). Given a subset S of the underlying set of a group G, its normalizer consists of all the elements from G such that the left and the right action by

those elements on S, results in the same subsets.

$$N(S) = N_G(S) = \{n \in G : Sn = nS\}$$

A normalizer forms a subgroup of G.

Definition 3.38 (Centralizer). Consider a group G, and a subset S of G, i.e., a subset of it's underlying set, then the centralizer subgroup of S in G is the subgroup of all the elements c in G which commute with all the elements of S.

$$C_G(S) = \{c \in G : cs = sc\}$$

Theorem 3.39. Given a subset $S \subset G$ of a group G, then the centralizer subgroup of S is a subgroup of the normalizer subgroup such that,

$$C_G(S) \subset N_G(S)$$

Definition 3.40 (Stabilizer). Consider a group G an let us define an action by the group G on a set X for every element $x \in X$ such that,

$$G \times X \to X$$

$$Stab_G(x) = \{g \in G : g \cdot x = x\} \tag{6}$$

Hence, a stabilizer subgroup of X consists of a set of elements under which all the elements of X are conjugation invariant.

3.41. Now, a stabilizer subgroup, also known as the **isotropy subgroup** of x is defined as the set of all the elements in G that leave x fixed. In other words, the action *stabilizes* every element of X, in other words.

Now, if all the stabilizer groups are trivial, then the action is called a **free action**.

3.42. Isotropy refers to a sense of uniformity in all orientations. In mathematical terms, it (G_x) is defined as exactly the above equation(6). Moreover, when two points x and y are on the same group orbit, say $y = gx$. then the isotropy groups are conjugate subgroups, that is, $G_y = gG_xg^{-1}$. More precisely, any subgroups conjugate to G_y occurs as an isotropy group G_y to some point y lying on the same orbit as x.

Theorem 3.43. Let $a \in G$, and $(G, \cdot)$ be a finite group, then the set generated by the power of a defined as: $S = \{a^n : n \in \mathbb{Z}, a \in G\}$ is finite.

Proof. Let $a \in G$. Then, $\cdots, a^{-3}, a^{-2}, a^{-1}, a^0, a^1, a^2, \cdots \in G$ where, $n \in \mathbb{Z}$. By closure property, the above holds. Now, $(G, \cdot)$ is a finite group, hence all the elements are not distinct. Let, $r < s : a^r = a^s$, then left action by a^{-r} gives us:

$$a^{-r} \cdot a^r = a^{-r} \cdot a^s \tag{7}$$

$$e = a^{-r} \cdot a^s = a^{s-r} \tag{8}$$

$$\therefore\ o(a) \leq (s - r) \tag{9}$$

The integral powers of a will result into non-distinct values. Hence, S is a finite set. $\square$

3.44. Let G, G' be monoids (sets with a law of composition which follows associativity and has an identity), then a **monoid-homomorphism** of G into G' is defined as a map $f : G \to G'$, such that $f(xy) = f(x)f(y)$ for all $x, y \in G$ and the identity in G is mapped to the identity in G'

3.45. If G and G' are groups, then the **group homomorphsim** between G and G' is simply the monoid homomorphism.

3.46. Let G and G' be groups and let $f : G \to G$" be a group homomorphism, then for all $x, y \in G$,

(1) $f(xy) = f(x)f(y)$
(2) $f(x^{-1}) = f(x)^{-1}$
(3) $e' = f(e) = f(xx^{-1}) = f(x)f(x^{-1}) = f(x)f(x)^{-1}$, where e' is the identity in G' and e is the identity in G.

The composition of two group homorphisms f and g, defined as $f \circ g$ or $g \circ f$ (if compatible,) is also a group homomrophism.

Note that for a cyclic group G, if $G =< x >$, then $G' =< f(x) > \Leftrightarrow f : G \to G'$ is a surjective homomorphism.

3.47. Let G, G' be groups, then a homomorphism $f : G \to G'$ is called an **isomorphism** if there exists a group homomorphism $g : G' \to G$ such that $f \circ g = 1_{G'}$ and $g \circ f = 1_G$, where 1_G and $1_{G'}$ represent **identity morphisms** on G and G', respectively.

Note that, it is not enough to say that the morphism is invertible for an isomorphism to exist, we need to ensure that the composition of the two morphisms gives the identity on the domain of the composition map.

For $f \circ g = 1_{G'}$, we need g to be an injective morphism and f to be a surjective morphism. Also, f is an isomorphism $\Leftrightarrow f$ is bijective.

An isomorphism ensures that two groups are same up to re-naming/representation of their elements. That is to say, under the isomorphism f, G and G' become identical, and the identity in G is mapped to the identity in G' and the order of the respective elements is also preserved.

3.48. If for a given isomorphism $f : G \to G'$, if $G = G'$, then f is said to be an **automorphism**, i.e., an isomorphism from the group G to itself.

Similarly, a homomorphism from the group G to itself is called an **endomorphism**.

Example 3.49. Consider the additive group $\mathbb{Z}/5\mathbb{Z} = \{0, 1, 2, 3, 4\}$ with the binary operation addition modulo 5. Let us define an endomorphism $f : \mathbb{Z}/5\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$ such that

$f(x) = x^x$. Then $f(0) = 0,\ f(1) = 1,\ f(2) = 4,\ f(3) = 2,\ f(4) = 1$. Here, $x \mapsto x^x$ is referred to as the $x-$**th power map**.

EXAMPLE 3.50. Let A be an abelian group. Define a map $f : A \to A$ such that $a \mapsto a^{-1}$. Then, clearly f is an endomorphism. And,

$$f(a_1a_2) = (a_1a_2)^{-1} = a_2^{-1}a_1^{-1} = a_1{-}1a_2^{-1} = f(a_1)f(a_2) = f(a_2)f(a_1) = f(a_2a_1)$$

is therefore a homomorphism. See $f(f(a)) = f(a^{-1}) = (a^{-1})^{-1} = a$. Therefore,

$$f \circ f = \mathrm{id}_A$$

and hence, f is injective and surjective. Moreover, $f^2 = \mathrm{id}$ also implies that f is an involution[4]. f is therefore an **automorphism**.

3.51. Consider the map $f : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$, such that $x \mapsto x(\mathrm{mod}\ 2)$. In this case, we can collect all those elements of $\mathbb{Z}$ which map to $0 \in \mathbb{Z}/2\mathbb{Z}$, the set of all such elements under the map f is referred to as the **kernel** of f, denoted ker f. Similarly, we can create a set of all those elements of $\mathbb{Z}$ which *do not* map to $0 \in \mathbb{Z}/2\mathbb{Z}$. We call this the **image** of f, denoted im f.

DEFINITION 3.52. (Kernel and Image). Let $f : G \to H$ be a group homomorphism. The **kernel** of f is defined as,

$$\mathrm{ker} f := \{a \in G \mid f(a) = e \in H\}$$

image of f is defined as,

$$\mathrm{im} f := \{b \in H \mid b = f(a)\ \forall a \in G\}$$

.

3.53. Let us define a group homomorphism $f : \mathbb{Z} \to 2\mathbb{Z}$, such that $x \mapsto 2x$. Then, only $0 \in \mathrm{ker} f$. In such a case, we say that the kernel is **trivial**. Clearly, f is also an injective morphism.

Whenever the kernel of a homomorphism is trivial, the map is injective.

EXAMPLE 3.54. Define a group homomorphism $f : G \to H$ such that $\mathrm{ker} f = \{e\}$, where e is the identity in G.

Consider $x, y \in G$ and let $f(x) = f(y)$, and multiply by $f(x^{-1})$ on the left:

$$f(x^{-1})f(x) = f(x^{-1})f(y) = f(x^{-1}y) = f(e) = e'$$

where e' is the identity in H. Hence, $x^{-1}y \in \mathrm{ker} f$, and $x^{-1}y = e$. Thus, $x = y$ and the map is injective.

3.55. If in 3.54, the f is also surjective, then f is an isomorphism.

[4]An involution is a function which is it's own self-inverse.

3.56. A homomorphism $f : G \to G'$ which establishes an isomorphism between G and its image in G', is referred to as an **embedding**. Thus, an injective homomorphism is an embedding, denoted as

$$f : G \hookrightarrow G'.$$

Definition 3.57. (Inverse Image.)

Let $f : G \to H$ be a group homomorphism. If B is a subset of H, then the inverse image of B under f is determined as

$$f^{-1}(B) := \{a \in G \mid f(a) \in B\}$$

Example 3.58. Let $f : G \to H$ be a group homomorphims. Then $\ker f$ is a subset of G. Let $g_1, g_2 \in \ker f$. Then,$f(g_1) = f(g_2) = e'$ in H. Then, for $g_1 g_2 = g_3$, we have

$$f(g_3) = f(g_1 g_2) = f(g_1)f(g_2) = e'$$

and hence $\ker f$ satisfies closure property. And we know $f(e) = e'$.

$$\Rightarrow f(g_1 g_1^{-1}) = e' = f(g_1)f(g_1^{-1}) = e'f(g_1^{-1}) = f(g_1^{-1}) = f(g_1)^{-1}$$

Therefore, $\ker f$ also consists of inverses. Hence, $\ker f$ *forms a subgroup of* G.

Definition 3.59. (Cosets.) Let G be a group and H a subgroup. A **left coset** of H in G is a subset of G, denoted as aH for some $a \in G$ such that

$$aH := \{ah, \ \forall \ h \in H\}.$$

Dually, a **right coset** of H in g is a subset of G, denoted as Hb for some $b \in G$ such that

$$Hb := \{hb, \ \forall \ b \in G\}.$$

3.60. Let H be a subgroup of a group G, and let aH be a left coset of H in G. Then the morphism $f : H \to aH$ such that $h \mapsto ah$ induces a bijection of H onto aH. Hence, any two left cosets (or right cosets) have the same cardinality.

Example 3.61. Let G be a group and let H be a subgroup of G and let aH and bH be two left cosets of H in G. Let aH and bH have one element in common then, the two cosets are the same.

Let $ah_1 = bh_2$, then $a = bh_2h_1^{-1}$

$$\Rightarrow aH = b(h_2h_1^{-1})H = bH$$

since for any $x \in H$, we have $xH = H$.

3.62. If G is a group, then it is the disjoint union of the left cosets of H, where $H < G$. Similarly, G can also be expressed as the disjoint union of the right cosets of H.

Definition 3.63. (Index.) The number of left or right cosets of a subgroup H in a group G is expressed as $(G : H)$ and is called the (left/ right) **index** of H in G.

3.64. By the definition of index above, the index of a trivial subgroup of a group G is $(G : 1)$ which is precisely the order of the group G.

EXAMPLE 3.65. Let $G = \mathbb{Z}$ be the group of integers under addition, and let $H = 2\mathbb{Z}$ be the subgroup consisting of even integers. Then $2\mathbb{Z}$ has two cosets in $\mathbb{Z}$ namely the set of even integers and the set of odd integers, so the index $|\mathbb{Z} : 2\mathbb{Z}|$ is 2.
More generally,

$$|\mathbb{Z} : n\mathbb{Z}| = n$$

for any positive integer n.

DEFINITION 3.66. (Normal Subgroup.) A subgroup H of a group G is called **normal** if every element of G *normalises* H, i.e., if $gHg^{-1} = H$ for all $g \in G$.

If H is a normal subgroup of G we denote it as $H \triangleleft G$.

3.67. Let $f : G \to H$ be a group homomorphism. Then the kernel of f is a normal subgroup of G.

3.68. If for each $g \in G$ and each $B \subset G$ let $g : B \mapsto gBg^{-1}$ where $gbg^{-1} = \{gbg^{-1} | b \in B\}$, then G is said to act on B by **conjugation**.

DEFINITION 3.69. (Group Action.) A **(left) group action** of a group G on a set A is a map $G \times A \to A$, for all $g \in G$ and $a \in A$, such that the following hold.

(1) $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$
(2) $e \cdot a = a$

Dually, **(right) group action** of a group G on a set A is a map $A \times G \to A$, for all $g \in G$ and $a \in A$, such that the above hold (respectively).

EXAMPLE 3.70. Consider the left group action defined as, $\sigma : G \times A \to A$ such that $\sigma_g(a) = g \cdot a$. Then, for all $a \in A$,

$$\begin{aligned}(\sigma_{g^{-1}} \circ \sigma_g)(a) &= \sigma_{g^{-1}}(\sigma_g(a)) \\ &= g^{-1} \cdot (g \cdot (a)) \\ &= (g^{-1}g) \cdot a \\ &= e \cdot a = a\end{aligned} \tag{10}$$

Hence, $(\sigma_{g^{-1}} \circ \sigma_g)$ is the identity map. Since g was arbitrary, $(\sigma_g \circ \sigma_{g^{-1}})$ is also the identity map. σ_g has a two-sided inverse.

Essentially, σ_g is a permutation of A.

Let us consider S_A to be the set of all the possible permutations of the elements of A. Define $\psi : G \to S_A$ such that $\psi(g) = \sigma_g$.

This map is well-defined, as $\sigma_g \in S_A$. Consider

$$\begin{aligned}\psi(g_1g_2)(a) &= \sigma_{g_1g_2}(a)\\ &= (g_1g_2)\cdot(a)\\ &= g_1\cdot(g_2\cdot a)\\ &= \sigma_{g_1}(\sigma_{g_2}(a))\\ &= (\psi(g_1)\circ\psi(g_2))(a)\end{aligned} \tag{11}$$

Therefore, ψ is a homomorphism.

3.71. The homomorphism discussed in 3.70 is called the **permutation representation** associated to the specified action.

Example 3.72. Let $ga = a$ for all $g \in G$, $a \in A$. This action is called the **trivial action** of the group G on A, and the associated permutation representation $G \to S_A$ is the *trivial* homomorphism, mapping every element of A to itself.

3.73. If for each $g \in G$ and each $B \subset G$ let $g : B \mapsto gBg^{-1}$ where $gbg^{-1} = \{gbg^{-1}|b \in B\}$, then G is said to act on B by **conjugation**.

3.74. If G acts on a set B non-trivially, such that different elements of G induce a distinct permutation of B, then G acts **faithfully** on B. Here, the associated permutation representation is *injective.*

Definition 3.75. (Kernel of a group action.) For the group action $G \times B \to B$, we can define the **kernel** to be

$$\{g \in G \mid gb = b,\ \forall\ b \in B\}$$

We say that the kernel consists of all the elements of G which *fix* all the elements of B.

3.76. For the trivial action, the kernel is the group G, and such an action is by definition, **unfaithful**.

That is, whenever $|\ker| > 1$, the group action is unfaithful.

3.77. By definition, a normal subgroup is conjugation invariant. Hence for a group G and it's normal subgroup H, we must have $xHx^{-1} = H$ for all $x \in G$.

Example 3.78. Let $f : G \to G'$ be a homomorphism and let $\ker f$ be it's kernel. From (**??**) we know that $\ker f < G$. Then for $x, y \in G$

$$x(\ker f) := \{xa \in G \mid f(a) = e' \in G'\}$$

and

$$(\ker f)y := \{ay \in G \mid f(a) = e' \in G'\}$$

Hence,

$$\begin{aligned} f(xa) &= f(x)f(a) \\ &= f(x)e' = f(x) \end{aligned} \tag{12}$$

and

$$\begin{aligned} f(ay) &= f(a)f(y) \\ &= e'f(y) = f(y) \end{aligned} \tag{13}$$

Therefore, $x(\ker f) = (\ker f)x \;\Rightarrow \ker f \triangleleft G$.

3.79. Given a homomorphism $f : G \to H$, $\ker f$ forms a **normal subgroup** of G.

Definition 3.80 (Quotient Group). Let G be a group and N be a normal subgroup of G, then the group G/N is called the **quotient group** or the **factor group**, defined as

$$G/N := \{gN \mid g \in G\}$$

G/H is read as G *modulo* H, or G *mod* H.

3.81. A quotient group can be viewed as the group of all cosets of a normal subgroup.

Definition 3.82 (Center). The centralizer of the group G is the center of the group G.

The **center** of G is the subgroup of G consisting of all elements of G which commute with all the other elements. The centre of a group is *normal*.

Example 3.83. Let us consider the multiplicative group of all invertible $n \times n$ (square) matrices with entries from $\mathbb{R}$, called the **General Linear Group**, written as $\mathrm{GL_n}(\mathbb{R})$.

Then the determinant is a homomorphism from $\mathrm{GL_n}(\mathbb{R})$ into $\mathbb{R} - \{0\}$.

$$\psi : \mathrm{GL_n}(\mathbb{R}) \to \mathbb{R} - \{0\}$$

Since all the elements of $\mathrm{GL_n}(\mathbb{R})$ are invertible, the determinant of all the matrices will be non-zero. Moreover, since the multiplicative structure has been considered in $\mathbb{R}$, we have $1 \in \mathbb{R}$ to be the identity. The kernel of ψ would consist of all those matrices whose determinant is 1. In this case, $\ker\psi$ is the $\mathrm{SL_n}(\mathbb{R})$, which is the **special linear group**[5].

Definition 3.84 (Automorphism Group). Let G be a group. The **automorphism group** of G, denoted $\mathrm{Aut}(G)$, is the set of all group isomorphisms from G to itself:

$$\mathrm{Aut}(G) := \{\varphi : G \to G \mid \varphi \text{ is an isomorphism}\}.$$

The group operation is given by composition of maps, i.e.,

$$(\varphi \circ \psi)(g) = \varphi(\psi(g)) \quad \text{for all } g \in G.$$

[5]The special linear group consists of all those elements of $\mathrm{GL_n}(\mathbb{R})$ whose determinant is 1.

EXAMPLE 3.85. Let us consider $\mathbb{R}^n$ as a vector space over $\mathbb{R}$. By definition, $\mathrm{Aut}(\mathbb{R}^n)$ consists of all bijective linear maps $T : \mathbb{R}^n \to \mathbb{R}^n$. Fix the standard basis $\{e_1, \ldots, e_n\}$ of $\mathbb{R}^n$. Then every linear transformation T is uniquely determined by its action on the basis and hence can be represented by a matrix $A \in M_n(\mathbb{R})$ such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$. Moreover, T is bijective iff A is invertible, which follows iff $\det(A) \neq 0$, i.e., $A \in \mathrm{GL}_n(\mathbb{R})$.

Define a map

$$\Phi : \mathrm{GL}_n(\mathbb{R}) \to \mathrm{Aut}(\mathbb{R}^n), \quad A \mapsto (x \mapsto Ax).$$

We claim that this map is a group homomorphism, since

$$\Phi(AB)(x) = ABx = A(Bx) = \Phi(A)(\Phi(B)(x))$$

so $\Phi(AB) = \Phi(A) \circ \Phi(B)$.

The map is injective because if $\Phi(A) = \Phi(B)$, then $Ax = Bx$ for all $x \in \mathbb{R}^n$, which implies $A = B$.

The map is surjective because every automorphism of $\mathbb{R}^n$ arises from an invertible linear transformation and hence from some matrix in $\mathrm{GL}_n(\mathbb{R})$. Therefore, Φ is a group isomorphism, and hence

$$\mathrm{Aut}(\mathbb{R}^n) \cong \mathrm{GL}_n(\mathbb{R}).$$

DEFINITION 3.86 (Inner Automorphism). Let G be a group. For each $g \in G$, define a map $\varphi_g : G \to G$ by $\varphi_g(x) = gxg^{-1}$. Such maps are called **inner automorphisms**. The set of all inner automorphisms is denoted by $\mathrm{Inn}(G)$:

$$\mathrm{Inn}(G) := \{\varphi_g \mid g \in G\}.$$

3.87. Note that each φ_g is an automorphism since it is bijective with inverse $\varphi_{g^{-1}}$, and it is also a homomorphism because $\varphi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\varphi_g(y)$.

THEOREM 3.88. $\mathrm{Inn}(G)$ is a group under composition.

PROOF. For $g, h \in G$, we have

$$\varphi_g \circ \varphi_h(x) = \varphi_g(hxh^{-1}) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = \varphi_{gh}(x), \quad \Rightarrow \varphi_g \circ \varphi_h = \varphi_{gh}$$

The identity element is φ_e, since $\varphi_e(x) = exe^{-1} = x$ for all $x \in G$.

The inverse of φ_g is $\varphi_{g^{-1}}$, since

$$\varphi_g \circ \varphi_{g^{-1}} = \varphi_e = \varphi_{g^{-1}} \circ \varphi_g.$$

Hence $\mathrm{Inn}(G)$ is a group under composition. □

DEFINITION 3.89 (Semidirect Product). Let K and H be groups, and let $\varphi : H \to \mathrm{Aut}(K)$ be a group homomorphism. The *semidirect product* of K by H (with respect to φ) is the set

$$K \rtimes_\varphi H := \{(k, h) \mid k \in K,\ h \in H\},$$

with the group operation defined by

$$(k_1, h_1) \cdot (k_2, h_2) := \big(k_1\, \varphi(h_1)(k_2),\ h_1 h_2\big).$$

Example 3.90. Let G be the set of all maps

$$T_{a,b} : \mathbb{R} \to \mathbb{R}, \quad T_{a,b}(x) = ax + b,\ a \neq 0$$

then G forms a group under composition of maps.

Let A be the multiplicative group of maps

$$T_{a,0}(x) = ax,\ a \neq 0$$

then,

$$A \cong \mathbb{R}^* = \mathbb{R} \backslash \{0\}.$$

Let B be the group of all translations.

$$T_{1,b}(x) = x + b$$

Then,

$$T_{a,b} \mapsto a$$

is a homomorphism of G onto the group A. And the kernel of this map will be the group B. Since B is normal in G, we observe that

$$G = AB = BA, \quad A \cap B = \{\mathrm{id}\}.$$

G is said to be the **semidirect product** of A and B.

3.91. A semidirect product is a generalization of a direct product. Whenever φ is trivial, we obtain the following definition of a direct product.

Definition 3.92 (Direct Product). Let K and H be groups. Their direct product is defined as

$$K \times H := \{(k, h) \mid k \in K,\ h \in H\},$$

with the group operation given by

$$(k_1, h_1)(k_2, h_2) = (k_1 k_2,\ h_1 h_2).$$

Example 3.93. Consider the direct product of $\mathbb{Z}/2\mathbb{Z}$ with itself.

$G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ consists of the following four elements:

$$G = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$$

where the group operation is component-wise addition modulo 2.

$(0, 0)$ is the identity element in G, and every non-identity element in has order 2.

3.94. $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is not isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

To determine if two groups of the same order are isomorphic, we examine the orders of their elements:

- In $\mathbb{Z}/4\mathbb{Z}$, there exists an element of order 4 (specifically, 1 and 3). Since there is an element that generates the entire group, $\mathbb{Z}/4\mathbb{Z}$ is cyclic.
- In $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, we calculate the order of each non-identity element:

$$(1,0) + (1,0) = (0,0)$$
$$(0,1) + (0,1) = (0,0)$$
$$(1,1) + (1,1) = (0,0)$$

Because there is no element of order 4, it cannot be isomorphic to $\mathbb{Z}/4\mathbb{Z}$. This group is known as the **Klein four-group**, briefly discussed in 3.23, denoted K_4 or V_4.

Example 3.95. Let us consider the semidirect product of $N = \mathbb{Z}/2\mathbb{Z}$ and $H = \mathbb{Z}/2\mathbb{Z}$.

The group $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ is a cyclic group of order 2. Any automorphism must map the generator 1 to another generator. Since 1 is the only generator, the only possible automorphism is the identity map:

$$\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z}) \cong \{id\}$$

Because the automorphism group is trivial, the only possible homomorphism $\phi : \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/2\mathbb{Z})$ is the trivial homomorphism (mapping every element to the identity).

When the action ϕ is trivial, the semidirect product is exactly the direct product:

$$\mathbb{Z}/2\mathbb{Z} \rtimes_{\text{triv}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \cong K_4$$

(26 Apr, 2026) This section on groups will be updated as and when required.

3. Rings

Definition 3.96 (Ring). A **ring** is a set R equipped with two binary operations, $+$ and $\cdot$, such that:

(1) $(R, +)$ is an abelian group.
(2) $(R, \cdot)$ is a semigroup.
(3) $a \cdot (b + c) = a \cdot b + a \cdot c$, $(a + b) \cdot c = a \cdot c + b \cdot c$, i.e., left and right distributivity holds.

Example 3.97. The set of integers ($\mathbb{Z}$), rationals ($\mathbb{Q}$), reals ($\mathbb{R}$), and complex numbers ($\mathbb{C}$) form a ring with respect to usual addition and multiplication.

Definition 3.98 (Unital Ring). A ring R is said to be **unital** if $(R, \cdot)$ forms a monoid, i.e., R has one unique identity element such that $1x = x$ for every $x \in R$.

3.99. Given a ring R, with additive and multiplicative identities 0 and 1, respectively. Then for all $a, b \in R$,

(1) $0 \cdot a = a \cdot 0 = 0$;
(2) $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$;
(3) $(-a) \cdot (-b) = a \cdot b$;
(4) $(na) \cdot b = a \cdot (nb) = n(a \cdot b)$ for all $n \in \mathbb{Z}$.

In the above, $na = a + \cdots + a$, where there are n copies of a in the sum.

Definition 3.100 (Commutative Ring). A ring R is said to be a **commutative** ring if for $x, y \in R$, we have $x \cdot y = y \cdot x$.

Example 3.101 (Zero Ring). A **zero ring** or trivial ring, is a unique ring (up to isomorphism) which consist of a single element, namely 0. The multiplicative and additive identites int his ring coincide.

Trivially, the zero ring is a commutative ring, where $0 \cdot 0 = 0$ and $0 + 0 = 0$.

Definition 3.102 (Ring homomorphism). A **ring homomorphism** is a structure preserving map defined as

$$f \colon A \to B \tag{14}$$

where A and B are rings such that

$$f(ab) = f(a)f(b), \quad f(a+b) = f(a) + f(b), \quad \text{and } f(1) = 1.$$

Therefore also $f(a - b) = f(a) - f(b)$.

Example 3.103. The most trivial example of homomorphism of rings is identity homomorphism $1_R \colon R \to R$ which is given by $1_R(r) = r$ for all $r \in R$.

Another example is mapping $f \colon \mathbb{Z} \to \mathbb{Q}$ by the map $f(n) = n$.

3.104. For a ring homomorphism $f \colon A \to B$ and another ring homomorphism $g \colon B \to A$, then there exist $f \circ g = 1_S$ where 1_S is **identity ring homomorphism** for ring S and $g \circ f = 1_R$ where 1_R is identity ring homomorphism for ring R, then f is called an **isomorphism**. The rings (R and S) are *isomorphic* and written $R \cong S$.

3.105. For two ring homomorphism f, g, there is another ring homomorphism h given by the composition $h = f \circ g$.

3.106. A ring isomorphism from R to itself is called **automorphism**.

Definition 3.107 (Kernel). A kernel (ker) of a ring homomorphism for $f \colon A \to B$ is defined as,

$$kerf \colon = \{r \in A | f(a) = 0\} \tag{15}$$

For two kernels $ker\ f$ and $ker\ g$ we have $ker\ f + ker\ g$ is also a kernel, as well as their composition.

Definition 3.108 (Subring). Given a ring homomorphism defined by $f\colon A \to B$ which is an inclusion of the sets and the operations are the same in A and B, then A is the **subring** of B.

Example 3.109. Consider the ring homomorphism $f : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$, where $\mathbb{Z}$ is the ring of integers and $\mathbb{Z}/3\mathbb{Z}$ is the ring of integers modulo 3. The kernel of f is the set of all integers n such that n is congruent to 0 (mod 3).
The kernel of f is the set $\{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$. This set is closed under addition and subtraction, it's also closed under multiplication by any element in $\mathbb{Z}$.

3.110. If we have a ring homomorphism $f\colon R \to S$ as an inclusion of set with the same operations, then R is the subset of S. So R is a ring, and a subring of S. Note that the operations on R and S are the same.

Proposition 3.111. A subset R is subring in S iff the following conditions are met

(1) Operations are closed under R. Every operation on S can be performed on R.
(2) R has an identity element $\mathbb{1}$.
(3) R is closed under additive inverse, so for every element r there exist a $\tilde{r}$ which is the additive inverse..

3.112. [Intersection of Rings] Any intersection of the collections of subrings of S is actually a subring.

If we have subrings for a ring S given which meet the conditions (1 to 3), then intersection of such rings are written as

$$R_I = R_1 \cap R_2 \cap R_3 \cap \cdots \cap R_n \tag{16}$$

3.113. [Subring Test] To check if R_I is the subring of the ring S, we check if the conditions of closure, identity and additive inverse are met. Since $a, b \in S$, a, b is in each of the subrings $R_1, R_2, R_3, \cdots, R_n$ as well, so R_I is closed under operations.

Since each of the subrings has identity and additive inverse, R_I has the identity element and additive inverse for the elements as well. Thus operations for S are also applicable to R_I. So R_I is a subring of S.

Definition 3.114 (Ideal). An (two-sided) **ideal** $\mathfrak{I}$ is a subgroup of the additive group $(R, +)$ which is closed under multiplication for the elements of R.

Equivalently, for any $a, b \in$:

$$a - b \in$$

and for any element $r \in R$ and any element $x \in I$:

$$rx \in I \quad \text{and} \quad xr \in I$$

3.115. A **left ideal** satisfies the additive subgroup condition and only requires $rx \in I$ for all $r \in R$ and $x \in I$.

Dually, a **right ideal** satisfies the additive subgroup condition and only requires $xr \in I$ for all $r \in R$ and $x \in I$.

3.116. If the ring R is commutative (where $ab = ba$ for all elements), then left ideals, right ideals, and two-sided ideals are all exactly the same thing.

3.117. An ideal may or may not contain an identity element; if it does, then the ideal is called **proper ideal**, otherwise called **trivial ideal**.

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